

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

1. **LO 4.4.A** — Express the motion of a point around a circle parametrically by applying transformations to $(x(t), y(t)) = (\cos t, \sin t)$, and parametrize the segment between two given points using an initial position and constant rates of change. _____

Vocabulary & Key Concepts

unit circle parametrization: The equations $x(t) = \cos t$, $y(t) = \sin t$, $0 \leq t \leq 2\pi$ model one full _____ revolution starting at _____ and centered at the _____.

radius / center: In $(h + r \cos t, k + r \sin t)$: r is the _____ and (h, k) is the _____ of the circular path.

orientation: Using $(\cos t, \sin t)$ gives _____ motion. Replacing t with $-t$ gives _____ motion.

linear parametrization: The parametrization $x(t) = x_1 + (x_2 - x_1)t$, $y(t) = y_1 + (y_2 - y_1)t$, $0 \leq t \leq 1$, traces the segment from _____ to _____.

Hook

Entry Question: A Ferris wheel has radius 30 ft with its center 35 ft above the ground. A rider boards at the rightmost seat.

Q1. What are the rider's starting coordinates (x, y) relative to a center at the origin? _____

Q2. As the wheel turns counterclockwise, what do you expect to happen to the rider's height y ? _____

LO 4.4.A — Part 1: Unit Circle Parametrization (EK 4.4.A.1)

Definition: The parametric function $(x(t), y(t)) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$, traces the _____ circle exactly _____ time, moving _____.

Complete the table:

t	$x(t) = \cos t$	$y(t) = \sin t$	(x, y)
0	_____	_____	_____
$\pi/2$	_____	_____	_____
π	_____	_____	_____
$3\pi/2$	_____	_____	_____
2π	_____	_____	_____

LO 4.4.A — Part 2: Transformations of the Circle (EK 4.4.A.2)

General form:

$$x(t) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} \cos t, \quad y(t) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} \sin t, \quad 0 \leq t \leq 2\pi$$

- Center: Radius:
- Counterclockwise when t increases / Clockwise: replace t with

Ferris Wheel Example. Radius 30 ft, center at $(0, 35)$, counterclockwise:

$$x(t) = \underline{\hspace{2cm}} \quad y(t) = \underline{\hspace{2cm}}$$

$$\text{Evaluate at } t = \pi/2: \quad x(\pi/2) = \underline{\hspace{2cm}}, \quad y(\pi/2) = \underline{\hspace{2cm}}$$

The rider is at position after a quarter turn.

Clockwise version (replace t with $-t$):

$$x(t) = \underline{\hspace{2cm}} \quad y(t) = \underline{\hspace{2cm}}$$

LO 4.4.A — Part 3: Linear Parametrization (EK 4.4.A.3)

Segment from (x_1, y_1) to (x_2, y_2) :

$$x(t) = x_1 + (x_2 - x_1)t, \quad y(t) = y_1 + (y_2 - y_1)t, \quad 0 \leq t \leq 1$$

At $t = 0$: position is . At $t = 1$: position is . At $t = 0.5$: position is the .

Example: Segment from $(2, 5)$ to $(6, 1)$.

$$x(t) = 2 + (\underline{\hspace{1cm}})t = \underline{\hspace{2cm}} \quad y(t) = 5 + (\underline{\hspace{1cm}})t = \underline{\hspace{2cm}}$$

Complete the table:

t	$x(t)$	$y(t)$	(x, y)
0	<u> </u>	<u> </u>	<u> </u>
0.5	<u> </u>	<u> </u>	<u> </u>
1	<u> </u>	<u> </u>	<u> </u>